

Precise pose measurement during RLV launch and landing using high-speed videogrammetry under uncontrolled outdoor lighting conditions

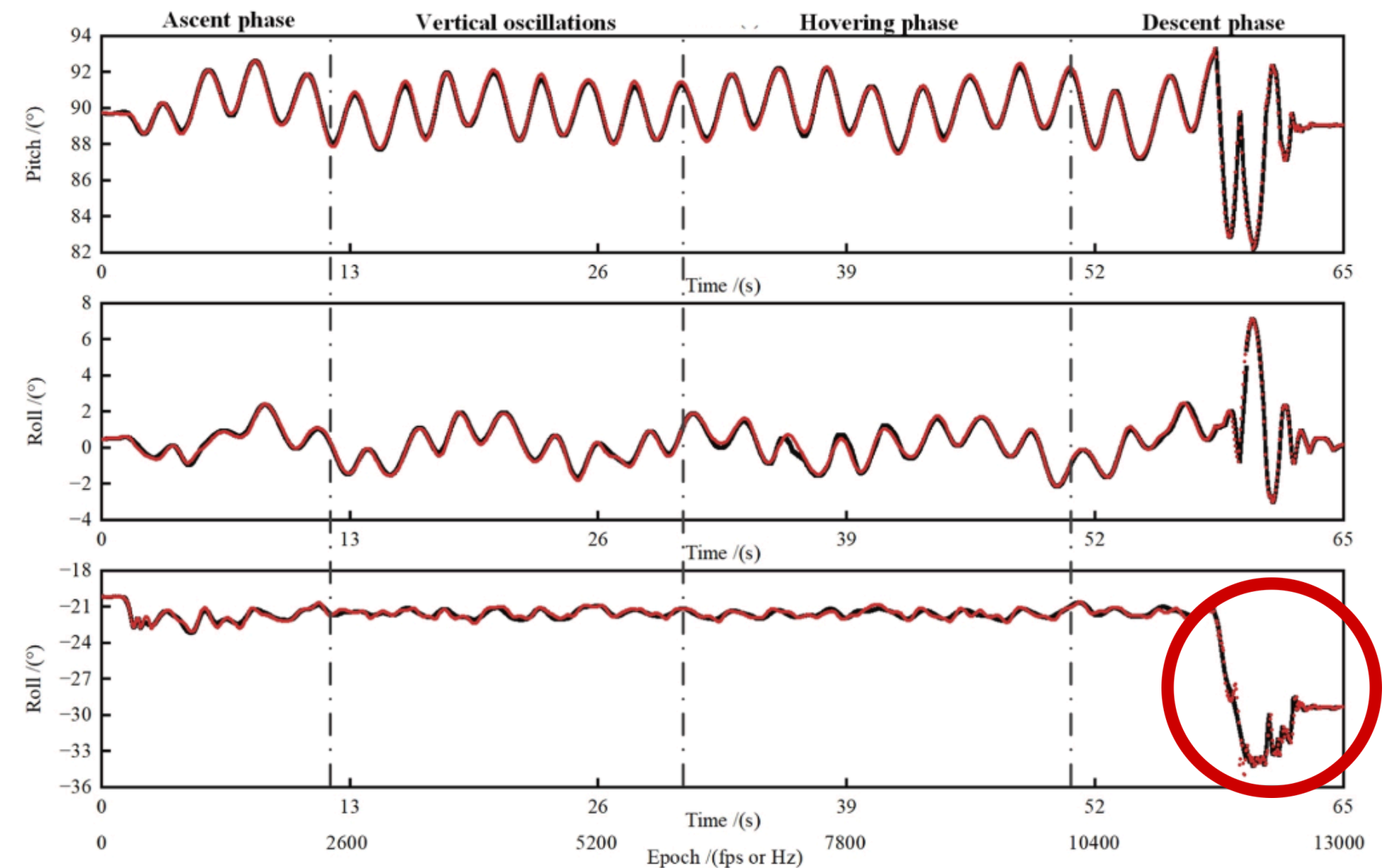
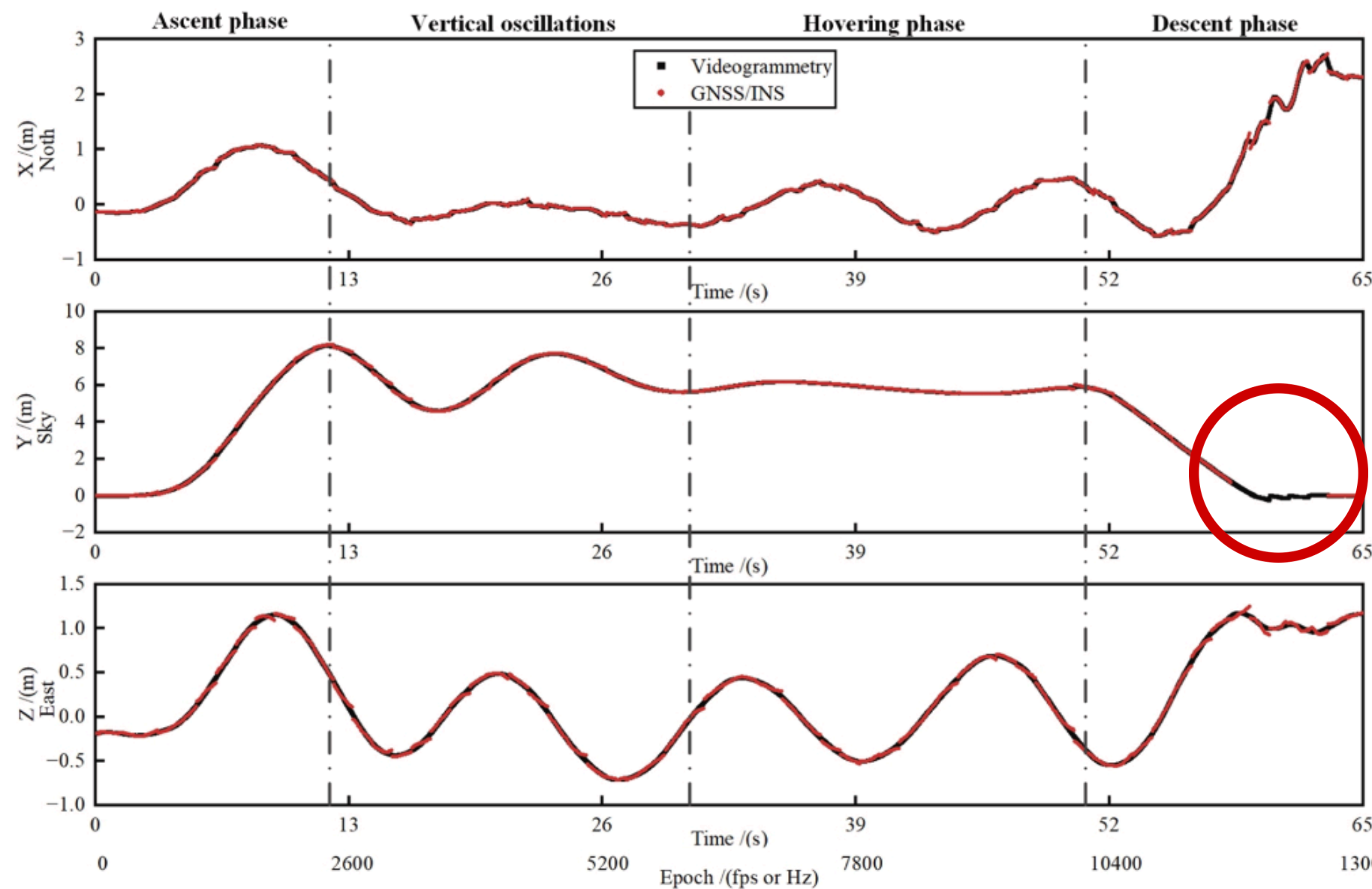
- Tong et al. (Jan, 2025)

Problems With On-Board Systems

INS systems are prone to **cumulative errors** (due to integration)

GNSS signals may become **weak or unavailable** in complex environments

Radar systems are susceptible to electromagnetic **interference**



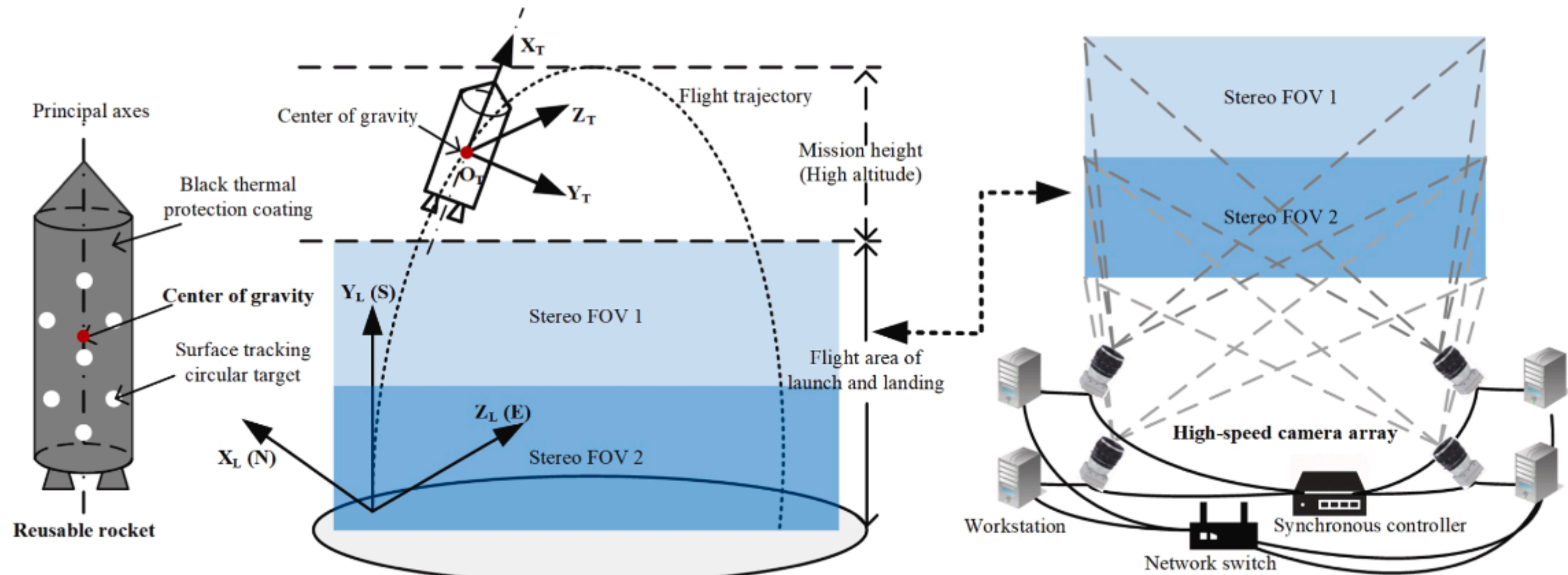
Why doing Stereo-based Pose Measurement Makes Sense

1. **Sub-millimeter 3D trajectory positioning** and **attitude accuracies of $\sim 0.2^\circ$** in roll, pitch, and yaw.
2. Provides **high-frequency (200 Hz) precise pose data** under uncontrolled outdoor lighting conditions
3. Making stereo pairs of cameras on ground allows a stereo FOV of the rocket.
4. All **computation is off-board** so no expensive computation in the rocket.
5. Completely reliable without any interferences.
6. **No heavy weight** put on the RLV.

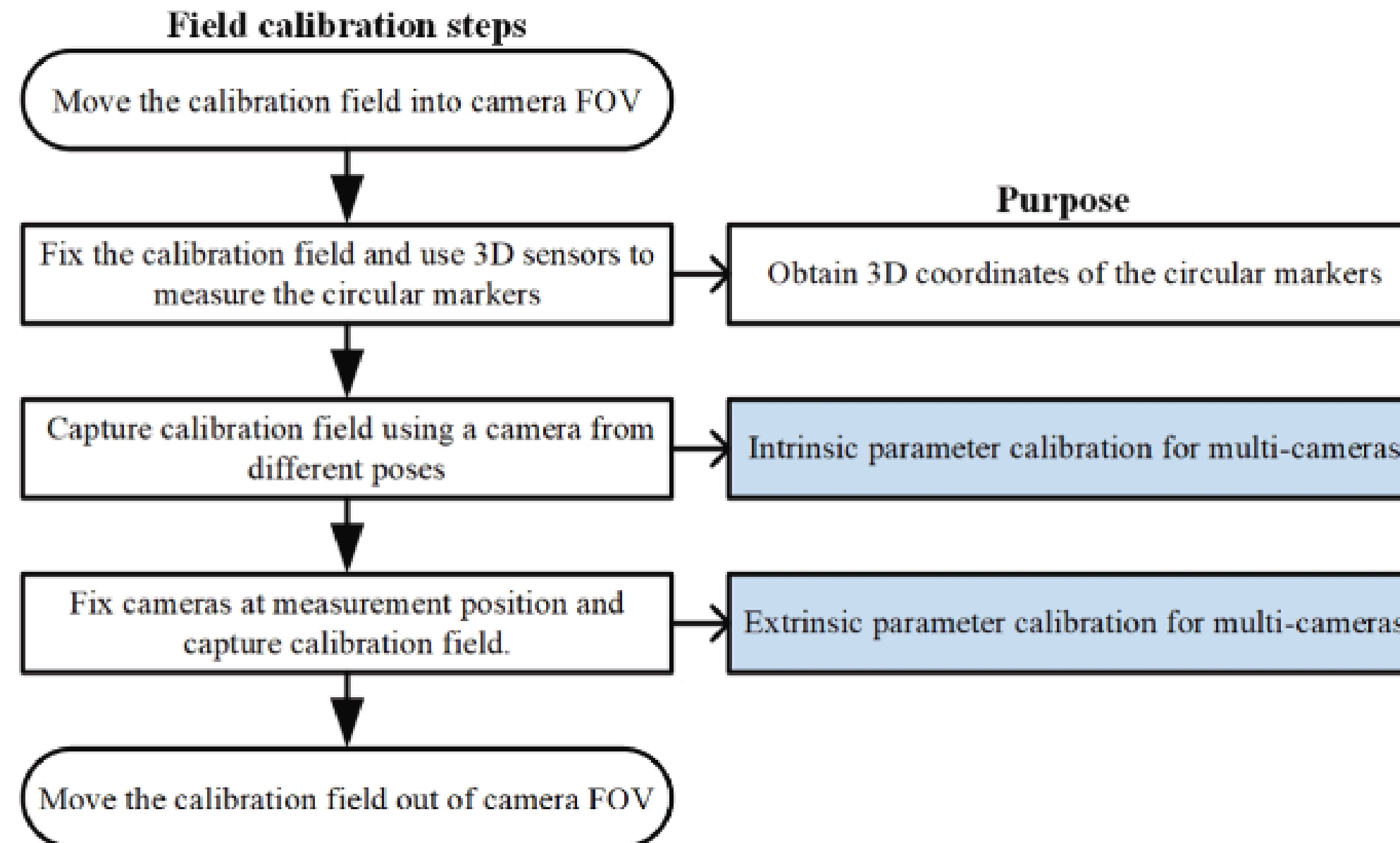
What the paper presents

1. A **precise** and **high-frequency** 6dof pose videogrammetric method for RLV during launch and landing.
2. A **flexible calibration strategy** used to accurately and efficiently determine the intrinsic and extrinsic parameters of all cameras.
3. A robust tracking algorithm based on **scale-adaptive image block segmentation and matching** is developed to continuously track non-coded circular targets on the rocket's surface, even under uncontrolled outdoor lighting conditions and variations in target size.

Methodology

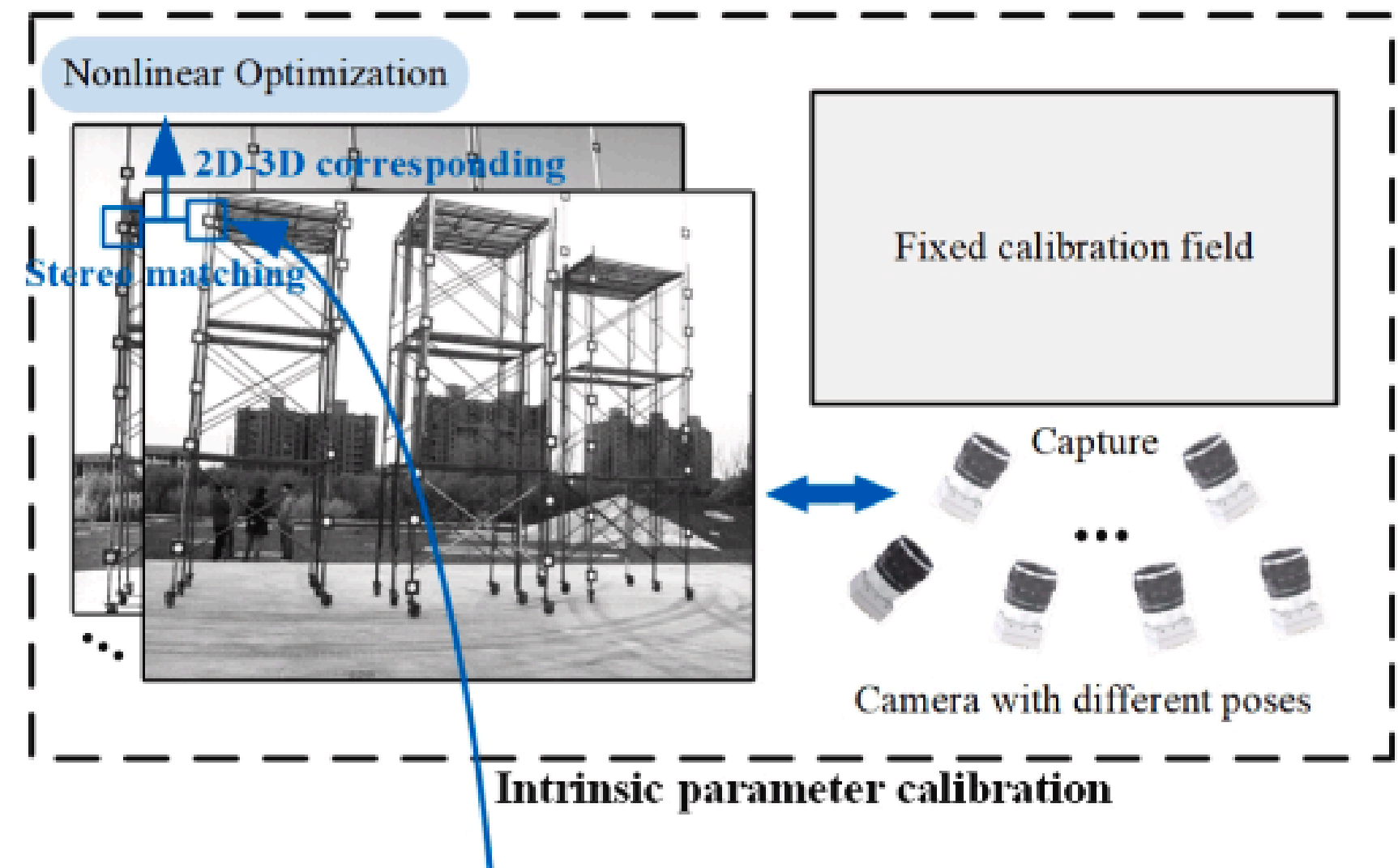


Calibration - Field Calibration



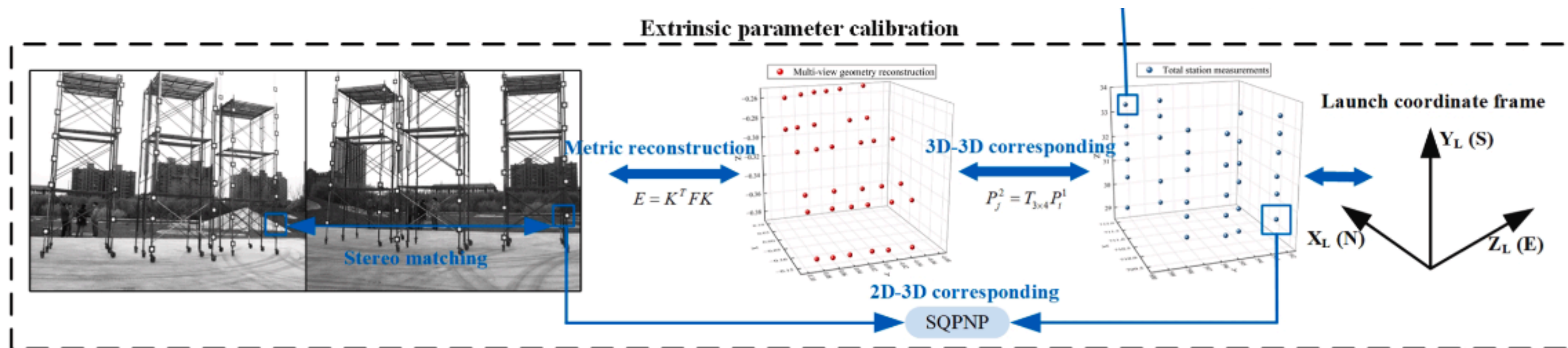
Calibration - Intrinsic Parameters Calibration

1. **Establish Correspondences:** Match 2D projections of the circular markers to their known 3D coordinates.
2. **Initial Estimation:** Minimize the reprojection error to derive initial estimates for focal length and relative extrinsic parameters.
3. **Final Refinement:** Account for lens distortion and apply nonlinear optimization to refine the intrinsic parameters.



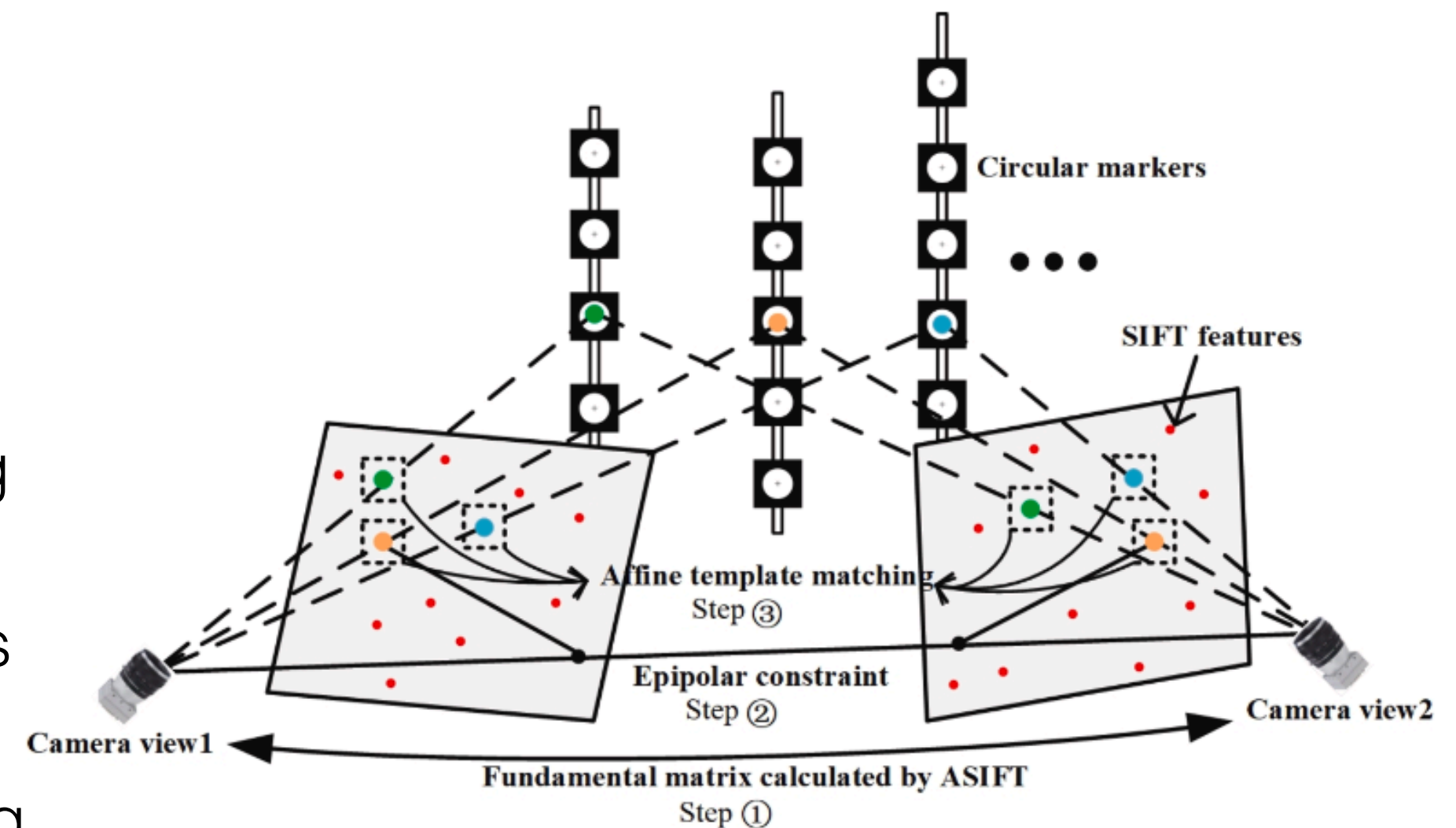
Calibration - Extrinsic Parameters Calibration

The 3D coordinates of the circular markers obtained by external sensors are transformed into the launch coordinate system. By associating these 3D coordinates with the corresponding 2D image coordinates, advanced Perspective-n-Point (PnP) algorithms are used to estimate the cameras' extrinsic parameter.



Calibration - Circular Target Matching

1. The fundamental matrix across images is calculated using the ASIFT feature-matching method.
2. The epipolar constraint is then derived from the fundamental matrix, and circular markers are matched across images using affine template matching combined with epipolar constraints.
3. Metric reconstruction of the markers is performed using multi-view geometry.
4. The fundamental matrix is refined using the matched circular markers.



Calibration - Circular Target Matching

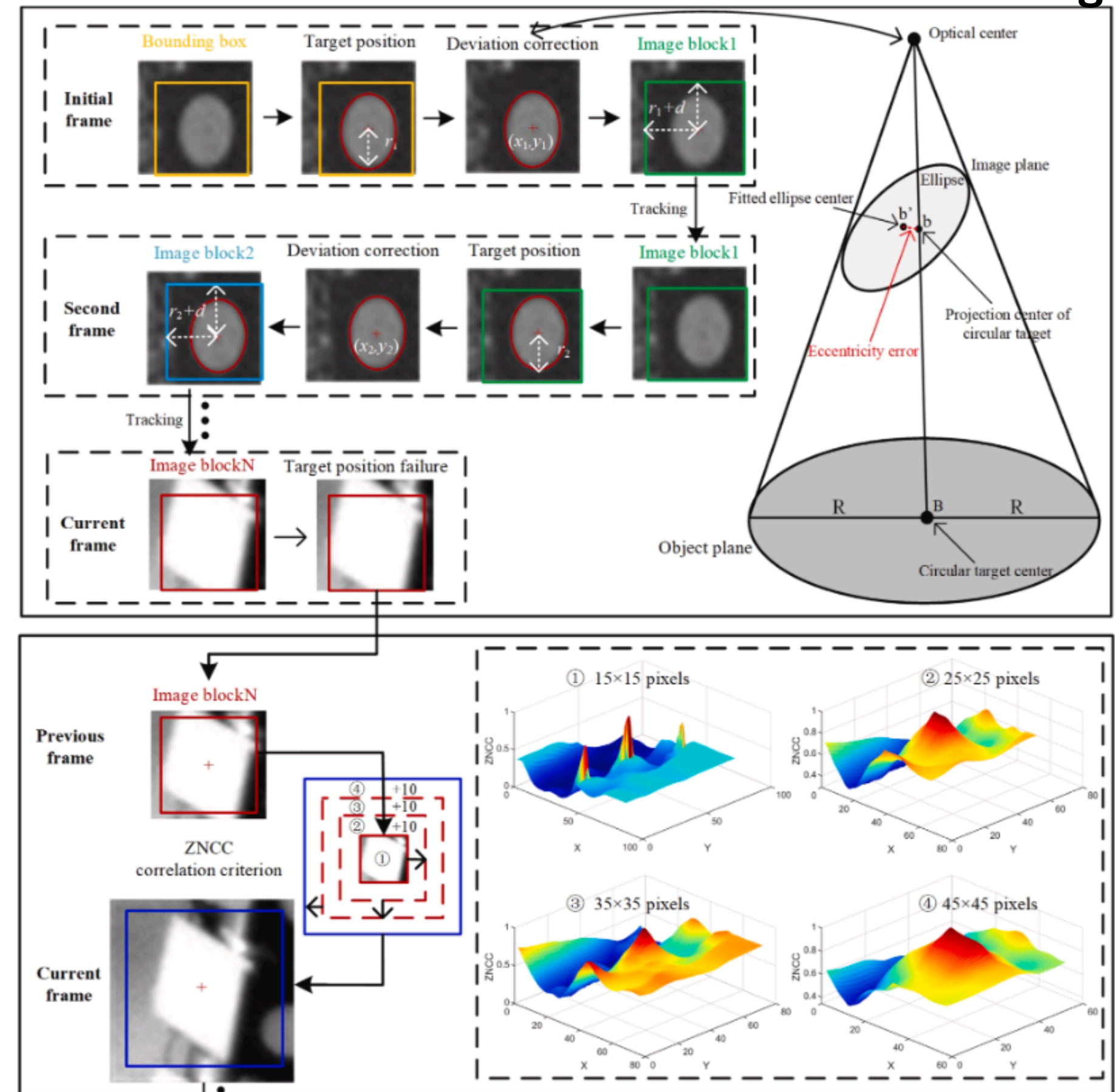
Epipolar constraints are a fancy way of relating points from a camera image to an image from another identical camera differing in pose.

The Epipolar lines pass through epipoles.



Methodology - Circular Target Tracking

- Step 1: Target recognition
- Step 2: Target position
- Step 3: Circular target center deviation correction
- Step 4: Scale-adaptive image block segmentation
- Step 5: Scale-adaptive image block matching



Methodology - Reconstruction of real trajectory and attitude

The rocket is modeled as a rigid body, meaning the spatial distances between surface targets remain constant throughout flight

Bundle Adjustment: Minimizing the sum of squared reprojection errors across all camera views and epochs.

Distance Constraints: Integrating rigid spatial relationships directly into the optimization objective.

$$\begin{cases} \min \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n (u_{ijk} - \pi(\text{Cam}_i, X_{jk}))^2 \\ \min(\|X_{jk} - X_{jk'}\| - D_{kk}), \forall k, k' \in 1, 2, \dots, n \end{cases},$$

